Linear Contexts, Tables, Equations, and Graphs

Answer Key

Grade 8 / Pre-Algebra

Quick Answers

1. 19, 5	4. -2	7. 2	10. 19, 3
2. 2	5. 7	8. 5	11. 4
3. 140	6. 36, -12	9. 39	12. 5, 8

1. Taxi fare table.

(a) Step 1. The question asks for a value the table already holds, so this is a read, not a computation: find the column headed 3 miles.
(a) Step 2. That column shows a fare of $\boxed{19}$ dollars.
(b) Step 1. Rate of change needs two rows of the table. Take the first and last columns, $(1, 9)$ and $(4, 24)$, so the arithmetic stays clean.
(b) Step 2. $\dfrac{24-9}{4-1} = \dfrac{15}{3} = \boxed{5}$ dollars per mile.
Check: every step across the table adds 5, and $9 - 5 = 4$ is the fixed flag-fall, so the rule is fare $= 5d + 4$.

2. Slope of Graph A.

Step 1. Slope needs two points that sit exactly on grid intersections, so read the y-intercept and one clean lattice point rather than estimating.
Step 2. The line passes through $(0, 3)$ and $(4, 11)$.
Step 3. $\dfrac{11-3}{4-0} = \dfrac{8}{4} = \boxed{2}$

3. Gym membership.

Step 1. The joining fee is paid once, so it is the initial value; the monthly charge repeats, so it is the rate. That fixes the shape $C = (\text{rate}) \cdot m + (\text{initial value})$.
Step 2. $C = 15m + 20$.
Step 3. Substitute $m = 8$: $\quad 15(8) + 20 = 120 + 20 = \boxed{140}$ dollars.

4. Slope of Graph B.

Step 1. The line falls from left to right, so expect a negative rate of change — that sign check catches the most common slip before it happens.
Step 2. The line passes through $(0, 12)$ and $(5, 2)$.
Step 3. $\dfrac{2-12}{5-0} = \dfrac{-10}{5} = \boxed{-2}$

5. Graph C: solve $4x = 28$.

Step 1. Here y is known and x is not, so the equation is used *backwards* — solve rather than substitute.

Step 2. $4x = 28$, so $x = \dfrac{28}{4}$.

Step 3. $x = \boxed{7}$

6. Draining tank.

(a) Step 1. The minute asked for appears in the table, so read the column headed 2 minutes.

(a) Step 2. The tank holds $\boxed{36}$ litres.

(b) Step 1. Water is leaving, so the rate of change must be negative; use the first and last columns, $(0, 60)$ and $(3, 24)$.

(b) Step 2. $\dfrac{24 - 60}{3 - 0} = \dfrac{-36}{3} = \boxed{-12}$ litres per minute.

7. Matching $y = 12 - 2x$ to a graph.

Step 1. An equation matches a graph when a point computed from the equation lies on that graph, so generate a point instead of eyeballing the picture.

Step 2. Substitute $x = 5$: $\quad 12 - 2(5) = 12 - 10 = \boxed{2}$.

Step 3. The equation therefore passes through $(5, 2)$. Graph B is the only reference graph containing that point — Graph A is rising and is at 13 when x is 5, and Graph C passes through the origin. So the equation matches **Graph B**. Its y-intercept, 12, and its falling direction agree as well.

8. Filling tank.

Step 1. Two (time, volume) pairs are given in words rather than in a table, but they are still two points of the same linear relationship: $(3, 26)$ and $(8, 51)$.

Step 2. $\dfrac{51 - 26}{8 - 3} = \dfrac{25}{5} = \boxed{5}$ litres per minute.

Check: the tank rises, so a positive rate is expected.

9. Seedling height.

Step 1. The present height is what you have before any weeks pass, so it is the initial value; the weekly growth repeats, so it is the rate.

Step 2. $h = 3w + 12$.

Step 3. Substitute $w = 9$: $\quad 3(9) + 12 = 27 + 12 = 39$, so the height is $\boxed{39\text{ cm}}$.

10. Table of a linear relationship.

(a) Step 1. The value is in the table, so read the column headed $x = 4$ rather than building an equation.

(a) Step 2. $y = \boxed{19}$.

(b) Step 1. Rate of change compares a change in y with the matching change in x; the columns step by 2, not by 1, so dividing is essential here.

(b) Step 2. Using the outer columns $(0, 7)$ and $(6, 25)$: $\dfrac{25 - 7}{6 - 0} = \dfrac{18}{6} = \boxed{3}$.

11. Comparing Graph A and Graph C.

Step 1. Both grids are identical, so the steeper line is the one with the larger rate of change; compute the unknown one and compare.

Step 2. Graph C runs through $(0, 0)$ and $(3, 12)$, so its slope is $\dfrac{12-0}{3-0} = \boxed{4}$.

Step 3. Graph A's slope is 2, so **Graph C climbs faster**. Without computing: on the same grid, Graph C rises four squares for every one square across while Graph A rises two, and Graph C reaches the top of the grid in half the horizontal distance.

12. Challenge: phone plan.

(a) Step 1. The two bills are two points of one linear relationship, $(2, 31)$ and $(5, 46)$, so the per-gigabyte charge is their rate of change.

(a) Step 2. $\dfrac{46-31}{5-2} = \dfrac{15}{3} = \boxed{5}$ dollars per gigabyte.

(b) Step 1. The fixed fee is the value at zero gigabytes. Work back from $(2, 31)$: two gigabytes cost $2 \times 5 = 10$, so the fee is $31 - 10 = 21$. The equation is $B = 5g + 21$.

(b) Step 2. A bill of 61 dollars means $5g + 21 = 61$, so $5g = 40$.

(b) Step 3. $g = \boxed{8}$ gigabytes.

Check: $5(8) + 21 = 61$.